STUDENT PROJECT REGISTRATION & MANAGEMENT PORTAL

Software Requirements & Presentation Document

Prepared for: Code Axis TechSolutions LLP

Purpose: A secure, CET Cell-style student portal for collecting academic/project requirements, generating quotations, receiving payments, managing development, and delivering completed college-project materials.

1. Project Vision

The proposed system is a professional student-facing web portal inspired by the structured workflow of an institutional registration portal such as a CET Cell website. Its purpose is to provide a single digital process through which students register, submit academic details and project requirements, upload college guidelines/reference files, receive a project quotation, complete secure payment, track development progress, and access final project deliverables.

2. Key Objectives

- Create a centralized student registration and project requirement system.
- Capture complete student, academic, college and project information.
- Allow students to upload college guidelines, documentation formats and ZIP/reference files.
- Enable the company team to review requirements and prepare project quotations.
- Provide secure online payment and automatic payment receipt generation.
- Track every project from registration to final delivery.
- Provide separate access for Students, Admins, Project Managers and Developers.
- Maintain strong security, document privacy, audit logs and backups.

3. Website Structure

Main Navigation	Purpose
Home	Portal introduction, services, announcements and registration/login actions
About Us	Company profile and project-development support information
Project Services	Available domains, technologies and service categories
How It Works	Registration → Requirement → Quotation → Payment → Development → Delivery
Student Registration	New student account/application creation
Student Login	Secure student dashboard access
Track Application	Application/project status tracking
Contact Us	Company contact and support information

4. Student Registration Module

Personal Details: Full Name, Mobile Number, Email, Date of Birth, Gender, Address, District, State and Student Photo.

Academic Details: Course, Branch, College, University, Current Year, Semester, Enrollment/PRN Number and Year of Passing.

Account Security: Password creation, OTP verification and secure authentication.

After successful registration, the system should generate a unique Application ID / Student ID, for example: **CAPS26-000125**.

5. Student Academic & Qualification Module

- Qualification: Diploma, BCA, BCS, BBA(CA), BSc, BE, BTech, MCA, MSc, MTech or Other.
- Branch / Specialization.
- College and University.
- Academic Year / Semester.
- Percentage / CGPA.
- Year of Passing.
- Option to add multiple qualification records where required.

6. Project Requirement Module

- Project title and project description.
- Whether the student already has a project topic: Yes / No.
- Request Project Suggestion option when no topic is available.
- Project domain: Web Development, Full Stack, Java, Python, Data Analytics, AI/ML, Android, Software Development, IoT and Other.
- Project type: Individual / Group.
- Group member details.
- Technology / programming language preference.
- Required modules and major features.
- Expected completion date.
- Additional requirements / special instructions.

7. College Guidelines & File Upload Module

Students must be able to upload the documents supplied by their college. Supported categories should include:

- College Project Guidelines
- Synopsis Format
- Documentation Format
- PPT Format
- Reference PDF / Images
- Existing Project Files
- ZIP File
- Other Supporting Documents

Files must be linked to the student's Application ID / Project ID and stored in private storage. ZIP files should be validated and malware-scanned before being made available to staff.

8. Project Review & Confirmation

Before final submission, students should see a complete preview of Personal Details, Academic Details, Project Details, Uploaded Documents and Additional Requirements. The student confirms that the information is correct and submits the project requirement.

9. Admin Review & Project Quotation

After submission, the Admin/Project Manager reviews the requirements and prepares a quotation based on project complexity, technology, modules, documentation and estimated delivery time.

Quotation Field	Required Information
Quotation Number	Unique quotation reference
Quotation Date / Valid Until	Issue date and validity period
Student / College	Applicant and academic information
Project	Project ID, title, type and technology
Services	Development, database, documentation, testing, deployment, etc.
Subtotal / Discount / Tax	Commercial calculation
Grand Total	Final payable amount
Payment Terms	Advance / milestone / final payment terms
Acceptance	Accept / Reject / Request Revision

10. Secure Payment Module

- Payment gateway integration for UPI, cards and net banking.
- Payment must be initiated only after quotation acceptance.
- Store gateway transaction reference, payment ID, amount, date and payment status.
- Generate downloadable payment receipt.
- Update project status automatically after verified payment.
- Never store raw card credentials or sensitive payment data in the application database.

11. Student Dashboard

The dashboard should show a clear project progress tracker:

Registration → Academic Details → Project Details → Documents → Quotation → Payment → Project Assigned → Development → Testing → Documentation → Final Delivery

- View/Edit permitted information
- View quotation and download PDF
- View payment status and receipt
- Upload additional documents
- View project progress
- Submit support queries
- Download final deliverables

12. Admin Dashboard

- Total Students / New Applications / Pending Review.
- Quotation Pending / Payment Pending / Active Projects / Completed Projects.
- Search and filters by College, University, Course, Branch, Domain, Payment Status and Project Status.
- Review student profile and uploaded files.
- Create, revise and send quotations.
- Approve / reject / request changes.
- Assign Project Manager and Developer.
- Monitor deadlines and project progress.
- View payment records and generate reports.

13. Developer / Project Team Panel

- View only assigned projects.
- Access approved project requirements and permitted documents.
- Update development status.
- Add technical notes and comments.
- Upload source code, database, documentation and final ZIP.
- Submit project for testing and final approval.

14. Project Status Workflow

New Registration → Requirement Submitted → Under Review → Quotation Prepared → Quotation Sent → Quotation Accepted → Payment Pending → Payment Completed → Project Assigned → Development Started → Development Completed → Testing → Documentation → Final Approval → Delivered → Completed

15. Security Requirements

- HTTPS / SSL throughout the portal.
- Password hashing using a strong password-hashing algorithm such as Argon2 or bcrypt.
- OTP-based email/mobile verification where appropriate.
- Role-Based Access Control (RBAC) for Student, Admin, Project Manager and Developer.
- Secure session management and rate limiting.
- Protection against SQL Injection, XSS, CSRF and common OWASP web vulnerabilities.
- Private document storage with authorization checks on every download.
- File extension, MIME type and size validation.
- ZIP validation and malware scanning.
- Audit trail for important administrative actions.
- Login activity and failed-login monitoring.
- Automated database backup and recovery plan.

16. Recommended Technology Architecture

Layer	Recommended Approach
Frontend	React / Next.js with responsive institutional-style UI
Backend	Node.js + Express or equivalent secure REST API
Database	PostgreSQL / MySQL
Authentication	Secure session/JWT + OTP verification
File Storage	Private object/file storage with access-controlled URLs
Payment	Razorpay / Cashfree / PhonePe Payment Gateway or equivalent
PDF	Server-side quotation and receipt PDF generation
Deployment	HTTPS-enabled production server with automated backups

17. Database – Core Entities

- Users / Students
- Qualifications
- Colleges / Universities
- Projects
- Project Members
- Project Requirements
- Documents / File Metadata
- Quotations / Quotation Items
- Payments / Transactions
- Project Assignments
- Project Status History
- Support Tickets / Messages
- Notifications
- Audit Logs

18. Final Deliverables

- Responsive public website.
- Student registration and login portal.
- Student dashboard.
- Admin dashboard.
- Project Manager / Developer panel.
- Quotation generation and PDF download.
- Secure online payment integration.
- Payment receipt generation.
- Project status tracking.
- Secure document and ZIP upload/download.
- Notification system.
- Reports and audit logs.

- Database backup and deployment documentation.

19. Business Workflow Summary

Student Registers → Submits Academic & Project Details → Uploads College Requirements → Company Reviews → Quotation Generated → Student Accepts → Secure Payment → Project Assigned → Development → Testing & Documentation → Final Project Delivery

20. Important Product Principle

The portal should be designed as a scalable institutional-style platform rather than a simple contact form. Every student record, project, quotation, payment and document should have a unique reference and an auditable status history. The interface should be simple for students while giving the company complete operational control from registration to final delivery.